Rakshith Shetty

AI Researcher & Engineer

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SUMMARY

AI researcher and engineer with a PhD and 10+ years applying machine learning across many domains. I take ideas from research concept to deployed product, with a track record of measurable real-world impact — **+$117M** in driven sales, defect-detection systems that improve over commercial vendors, and models processing data for **500M+ customers**. My experience spans embedded signal processing, computer vision, generative AI, and large-scale production systems.

EXPERIENCE

Applied Scientist — 2021–Present

Amazon | Berlin

- **Customer Intelligence at Scale**
  - Built a high-scale customer-understanding suite combining LLMs and tabular models.
  - Processed behavioral data for **500MM customers** and **350MM queries monthly**, driving **+$117MM annualized sales**.
  - Curated datasets up to **45B records** (SQL) and deployed scalable inference on AWS (Fargate, SageMaker).
- **High-Precision Visual Inspection (Pharmacy QC)**
  - Engineered a production computer-vision inspection system for pharmaceutical quality control.
  - Trained a Detectron2 detector with a custom NMS algorithm tuned for extremely low missed-alarm rates.
  - **91% reduction in false alarms** (11% to 1%) and **98.5% auto-pass**, outperforming a licensed third-party vendor.
- **Multimodal Pipeline for Automatic Video Creation**
  - Built an end-to-end system that generates product showcase videos from existing image and video assets.
  - Pipeline: open-vocabulary detection (OwlViT) + OCR + an LLM agent (Claude) for content planning, narration and orchestration.
  - Shipped **11k+ videos** with a **+0.18% conversion lift**.
- **Large-Scale Image Classification & Feature Learning**
  - Trained image/video/text models recognizing nine complex visual attributes at **90% precision**.
  - Used self-supervised contrastive learning and hard-negative mining for long-tail, in-the-wild data.
  - Cut annotation turnaround **24x** (4 days to 4 hours) and enabled a **5x** increase in data volume.

PhD Researcher, Generative Modeling & Robustness — 2017–2020

Max Planck Institute for Informatics | Saarbrücken

- **Controllable Generation & Model Robustness**
  - Built generative models for controllable synthesis and editing of images and text.
  - Used synthetic data generation to probe and improve robustness of vision models
  - Output: 7 papers in top conferences, thesis awarded Summa cum Laude.

Systems Design Engineer — 2011–2014

Broadcom India Research | Bangalore

- **Embedded Signal Processing & Hardware**
  - Developed real-time physical-layer DSP firmware in C for WLAN and 4G/LTE chipsets under tight hardware constraints.
  - Validated hardware-software alignment via MATLAB simulation and silicon bring-up; promoted to Staff Engineer in year two.

EDUCATION

PhD, Computer Vision & Machine Learning, *Summa Cum Laude* — Max Planck Institute for Informatics
Thesis: Adversarial Content Manipulation for Analyzing and Improving Model Robustness. **2017–2020**

MSc, Machine Learning & Data Mining — Aalto University, Helsinki (4.84/5.0) **2014–2016**

BEng, Electronics & Communication — Visvesvaraya Technological University (9.04/10.0) **2007–2011**

ACHIEVEMENTS & HONORS

- **Qualcomm Innovation Fellowship, Europe** (2018) — one of three research proposals funded ($40,000) across all of Europe.
- **Three awards for Master's Thesis** (2017) — Finnish Society for Computer Science, Aalto School of Science, Media Industry Research Foundation of Finland.
- **Joint Winner**, Microsoft Video-to-Language Challenge, ACMMM 2016; **Winner**, Large Scale Movie Description Challenge, ICCV 2015.
- **Reviewer at top AI conferences:** CVPR, NeurIPS, ICCV, ECCV, ICML, ICLR, AAAI (2018–present).

SELECTED PUBLICATIONS

1,400+ citations — full list on Google Scholar

V. Agarwal, **R. Shetty**, and M. Fritz, Towards causal VQA: Revealing and reducing spurious correlations by invariant and covariant semantic editing, *CVPR*, Jun. 2020.

R. Shetty, M. Fritz, and B. Schiele, Towards automated testing and robustification by semantic adversarial data generation, *ECCV (oral)*, Aug. 2020.

R. Shetty, B. Schiele, and M. Fritz, Not using the car to see the sidewalk: Quantifying and controlling the effects of context in classification and segmentation, *CVPR*, 2019.

R. Shetty, M. Fritz, and B. Schiele, Adversarial scene editing: Automatic object removal from weak supervision, *NeurIPS*, 2018.

R. Shetty, B. Schiele, and M. Fritz, A^4NT: Author attribute anonymity by adversarial training of neural machine translation, *USENIX*, 2018.

R. Shetty, M. Rohrbach, L. A. Hendricks, M. Fritz, and B. Schiele, Speaking the same language: Matching machine to human captions by adversarial training, *ICCV*, 2017.

H. R. Tavakoli, **R. Shetty**, A. Borji, and J. Laaksonen, Paying attention to descriptions generated by image captioning models, *ICCV*, 2017.

SKILLS

Domains: Computer Vision, NLP & LLMs, Generative AI, Signal Processing, Industrial Inspection, Behavioral Modeling
Machine Learning: PyTorch, Generative Models, Self-Supervised & Contrastive Learning, Distributed/Multi-GPU Training
Engineering: Python, C/C++, SQL, AWS, Embedded/Real-Time Systems
Foundations: Linear Algebra, Probability & Statistics, Optimization

LANGUAGES

English (Proficient), Kannada (Native), Hindi (Intermediate), German (B1)